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RTCA, Inc.

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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 1

Purpose and Applicability

Section 2

Definitions of Terms

Section 3

Conditions of Tests

Important Notice

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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1 Purpose and Applicability

This document defines a series of minimum standard environmental test conditions (categories) and applicable test procedures for airborne equipment. The purpose of these tests is to provide a laboratory means of determining the performance characteristics of airborne equipment in environmental conditions representative of those which may be encountered in airborne operation of the equipment.

The standard environmental test conditions and test procedures contained herein may be used in conjunction with applicable equipment performance standards as a minimum specification under environmental conditions, which can ensure a sufficient degree of confidence in performance during operations.

Note: In each of the test procedures contained herein, the following phrase will be seen several times:

DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

The "applicable equipment performance standards" referred to are either:

a. EUROCAE Minimum Operational Performance Specifications (formerly Requirements) (MOPS/MOPR).

b. RTCA Minimum Performance Standards (MPS) and/or RTCA Minimum Operational Performance Standards (MOPS).

c. The manufacturers equipment specification(s), where applicable.

Some of the environmental conditions and test procedures contained in this document are not necessarily applicable to all airborne equipment. The selection of the appropriate and/or additional environmental conditions and test procedures is the responsibility of the writers (authors) of the performance standards for the specific airborne equipment.

Notes:

1. There are several additional environmental conditions (categories), that specific airborne equipment may need to be evaluated against, that have not been included in this document. These include, but are not limited to: hail, acceleration and acoustic vibration.

2. The procedures for testing airborne equipment for special environmental conditions that are usually uniquely related to that specific type of airborne equipment, should be the responsibility of the writer (author) of the performance standard for that specific equipment.

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3. The International System of Units (SI) is usually used throughout this document as the primary values. In certain instances, however, when the primary values were derived in US units, these units are used as the primary values.

4. Subject to the provisions of Subsection 3.2, it is permissible to use more than one test article.

The words airborne equipment, as used within this document, have direct applicability to most airborne equipment. It is the responsibility of those who wish to apply the test conditions and procedures contained in this document to determine the applicability of these test conditions and procedures to a specific equipment intended for installation on, or within, a specific or general class or type of aircraft.

Minimum operational performance standards (MOPS) prepared by RTCA, Inc. for airborne equipment contain requirements that the equipment must meet to ensure reliable operation in actual aeronautical installations. These equipment requirements must be verified in ambient and stressed environmental conditions. The MOPS typically contain recommended bench test procedures for ambient conditions, and refer to RTCA Document DO-160, "Environmental Conditions and Test Procedures for Airborne Equipment," for the stressed environmental testing. The test categories defined in DO-160 are intended to encompass the full spectrum of environmental conditions that airborne equipment may experience – from benign to very hostile.

The environmental conditions and test procedures defined herein are intended to determine only the performance of the airborne equipment under these environmental conditions and are not intended to be used as a measure of service life of the airborne equipment subjected to these tests.

Any regulatory application of this document is the sole responsibility of appropriate governmental (regulatory) agencies.

1.1 Historical Note, and General Guidance to Users

DO-160 (or its precursor, DO-138) has been used as a standard for environmental qualification testing since 1958. It has been referenced in Minimum Operational Performance Specifications (MOPS) for specific equipment designs, and is referenced in FAA Advisory Circulars as a means of environmental qualification for TSO authorization. It has been subject to a continuous process of upgrade and revision as new needs arose within the aviation community, as improved test techniques have emerged, and as the realities of equipment operation of under actual environmental conditions has become better understood.

Environmental stresses can result from natural forces or man made effects, and may be mitigated by details of the equipment installation. The categories that have been

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developed over time reflect a reasonably mature understanding as to the severity of the stresses, the degrees of mitigation achievable in the design of an installation, and the robustness that must be designed into equipment in order to perform under the resultant stress. In order to fully reap the advantages of the maturity of this document, it is incumbent upon the designers of the installed equipment, as well as the designers of the host installation, to consider the categories defined herein as early in their programs as feasible. The categories defined within each environmental test procedure have proved to be a practical set of boundary conditions between the requirements of real world installations and the performance of installed equipment. Effective dialogue between the airframe and equipment designers is essential to ensure that correct categories are utilized.

1.2 DO-160 User Guides

Various sections within this document now contain User Guide material. This material appears at the end of the respective section. The User Guides provide background information for the associated test procedures and requirements in the main body of DO-160. This information includes rationale for requirements, guidance in applying the requirements, commentary, possible trouble shooting techniques and lessons learned from previous laboratory experience. This information is intended to help users understand the intent behind the requirements, and to help users develop detailed test procedures based on the general test procedures in this document. These User Guides are provided for guidance purposes. The User Guides do not contain requirements. Only the procedures in the body of each section are required to be applied to the EUT.

The User Guides follow the same general format as the main body of the section. Main body paragraph numbers are repeated in the User Guide.

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2 Definitions of Terms, General

This section contains the definitions of general terms that are utilized throughout this document. The definition of terms specific to a particular section may be found in the appropriate section.

2.1 Equipment Temperature Stabilization

a. Not Operating

The equipment is considered temperature stabilized when the temperature of the functioning parts of the test item considered to have the longest thermal lag are within three degrees Celsius of the specified test temperature. When temperature measurement of the largest internal mass is not practical, the minimum time considered applicable for temperature stabilization shall be three hours.

b. Operating

The equipment is considered temperature stabilized when the functioning parts of the test item considered to have the longest thermal lag do not vary by more than two degrees Celsius per hour. When temperature measurement of the largest internal mass is not practical, the minimum time considered applicable for temperature stabilization shall be two hours.

2.2 Maximum Duty Cycle

When operation of an equipment is periodic, the maximum duty cycle is the relationship between the maximum length of time for which the equipment is designed to operate at its rated capacity and the length of time during which the equipment is not operating; or when the operating capacity is at a defined minimum. The maximum duty cycle shall be established by the equipment specification.

2.3 Not Operating

Not operating is that condition wherein no power is applied to the equipment unless otherwise defined in the individual equipment specification.

2.4 Controlled or Partially Controlled Temperature Locations

A controlled or partially controlled temperature location is a space within an aircraft in which the temperature of the air is maintained by an environmental control system (see Table 4-1 of applicable category).

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2.5 Total Excursion

Total excursion means the total displacement from positive maximum to negative maximum.

2.6 Equipment

The term "equipment" includes the test items and all of the components or units necessary (as determined by the equipment manufacturer) for the equipment to properly perform its intended function(s). The equipment shall be representative of the production standard that will be utilized in service.

2.7 Altitude

Altitude represents the environmental pressure relative to sea level to which the equipment is exposed during the tests.

2.8 Category of Tests and Declarations

For each environmental condition addressed in this document, the equipment supplier should select, from categories defined within the particular sections, a category which best represents the most severe environment which the equipment is expected to be regularly exposed to during its service life. An exception to this rule is the use of Category X as explained below. The selected categories selections are to be tabulated on the Environmental Qualification Form in accordance with the guidelines presented in Appendix A.

For any category listed on the Environmental Qualification form it can be derived that the equipment would also be able to perform its intended function(s) when exposed to categories that can be assessed as being less severe.

Use of Category X on the environmental qualification form in association with any environmental test procedure of this document is reserved for the case where the equipment supplier wishes to indicate that compliance with equipment performance standards has not been demonstrated under the environmental conditions addressed by that particular procedure.

If any category listed on the Environmental Qualification Form is not sufficient for a specific aircraft certification, then additional testing may be required.

When the statement "DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS" is found at the end of or during the test procedures, it should be understood that performance compliance and verification is considered to be the requirement that allows the equipment to be certified as to its ability to perform its intended function(s) during and/or after a specific test category.

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2.9 Applicability of Test Results

The results of the testing using these procedures are only valid for the test configuration (test set up, external configuration and internal configuration) used during the qualification tests. Any change from this configuration, either externally or internally (such as: PCB lay-out, component change inside the unit, installation wiring, etc.) must be assessed to ensure that the test results are still applicable. If an assessment cannot show that the results are still applicable, re-testing is required.

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3 Conditions of Tests

3.1 Connection and Orientation of Equipment

Unless otherwise stated, connect and orient the equipment (e.g. mechanically and electrically) as recommended by the equipment manufacturer for normal service installation, including any cooling provisions, as necessary to perform the tests and to DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS. Interconnecting cable lengths, where not specified, shall be at least 1.50 m long and shall be configured so as to allow one common bundle of 1.20 m. Any inputs or outputs to or from other equipment(s) normally associated with the equipment(s) under test shall be connected or adequately simulated.

Note: Paragraphs 19.3 and 20.3, if applicable, will require an interconnecting cable longer than these minimums.

3.2 Order of Tests, Multiple Test Articles

It is the responsibility of the equipment manufacturer to determine any requirements for cumulative or combined testing, and to reflect these requirements in the equipment specification and test plan. Insofar as any such requirements are not compromised, multiple test articles may be used, tests may be performed in any order, and separate test articles may be used for demonstrating compliance with the separate tests.

Note: In cases where cumulative testing is required, or multiple tests are to be performed upon a single test article, the following provisions shall apply:

a. The salt fog test shall not be conducted prior to the fungus resistance test.

b. The sand and dust test shall not be conducted prior to the fungus resistance, salt fog, or humidity tests.

c. The explosive atmosphere tests shall not be conducted prior to any other DO-160 tests.

d. The flammability tests shall not be conducted prior to any other DO-160 tests.

3.3 Combining Tests

It is acceptable to employ alternate procedures developed as combinations of the procedures described herein, provided it can be demonstrated that all applicable environmental conditions specified in the original procedures are duplicated or exceeded in the combined procedure. If alternate procedures are used, appropriate information should be provided along with the Environmental Qualification Form, (see Appendix A).

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3.4 Measurement of Air Temperature in the Test Chamber

The temperature of the air in the test chamber shall be measured at a location where the air conditions are representative of that immediately surrounding the equipment. Measurement of chamber wall temperature is not suitable, due to temperature lag and heat transfer through the chamber wall.

A means of circulating air in the test chamber should be employed to assure an approximate uniform air temperature condition throughout the chamber. When such means are employed, the air movement shall not be directed on the equipment under test, and the equipment shall be tested at the minimum flow rate consistent with the purpose of maintaining a uniform temperature distribution in the chamber. The ambient air velocities surrounding equipment not requiring auxiliary cooling shall remain comparable to those air velocities that occur from natural convection.

For equipment that requires auxiliary cooling to assure proper operation, as defined in the equipment installation instructions, the following applies:

a. If air is the cooling medium, the supplied cooling air characteristics shall be the same as the specified chamber air characteristics, unless otherwise specified by the equipment manufacturer.

b. If the cooling medium is not air, the medium and its supply temperature shall be as specified by the equipment manufacturer.

Note: For equipment whose installation location is known and defined relative to other equipment, sources of radiated heat and/or impediments to normal convection should be simulated in the test.

3.5 Ambient Conditions

Unless otherwise specified, all tests shall be made within the following ambient conditions:

a. Temperature: +15 to +35 degrees Celsius.

b. Relative Humidity: Not greater than 85 percent.

c. Ambient Pressure: 84 to 107 kPa (equivalent to +5,000 to -1,500 ft) (+1525 to -460 m).

When tests are conducted at ambient conditions that differ from the above values, the actual conditions shall be recorded.

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3.6 Environmental Test Condition Tolerances

Unless otherwise specified, tests made at environmental conditions other than ambient, as defined above, shall be conducted subject to the following tolerances:

a. Temperature: ⟨≡~ 3 degrees Celsius.

b. Altitude: ⟨≡~ 5 percent of specified pressure.

3.7 Test Equipment

All stimulus and measurement equipment used in the performance of the tests should be identified by make, model, serial number and the calibration expiration date and/or the valid period of calibration where appropriate. When appropriate, all test equipment calibration standards should be traceable to national and/or international standards.

3.8 Multiple Unit Equipment

If the equipment to be tested consists of several separate units, these units may be tested separately, provided the functional aspects are maintained as defined in the relevant equipment specification.

3.9 EUT Configuration for Susceptibility Tests

During any environmental test of electrical and electronic equipment (for example, susceptibility test) conducted per this document, the equipment under test shall be configured in the most sensitive functioning mode that could be encountered during its normal operation in the field. When the equipment is embedded with basic and/or application software, it must be tested with software functions, operating (or simulated), to exhibit the maximum sensitivity to the test environment. The description and the justification for the functional configuration, including software, shall be reported or referenced in the test report.

Unless otherwise specified in the applicable minimum performance standard, special purpose test firmware/software is acceptable provided that the hardware and interfaces are comprehensively exercised and yields good test coverage based on validated requirements, and the configuration is controlled. The special purpose test firmware/software, when used, should be capable of thoroughly exercising the hardware functions.

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